Online appendix for "Disclosure Regulation, Intangible Capital, and the Disappearance of Public Firms"

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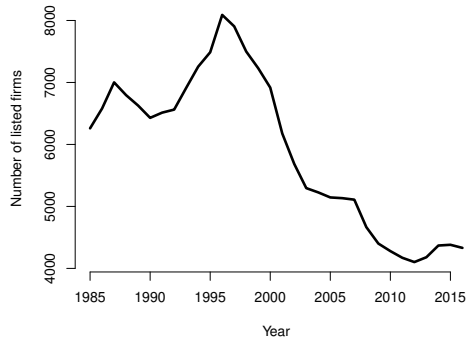
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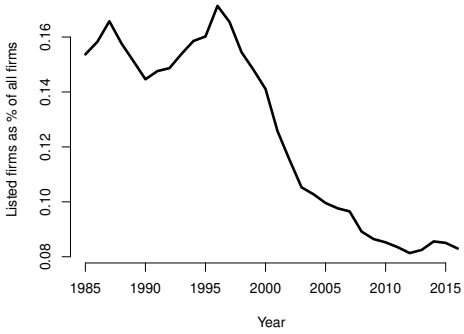
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A Data and measurement

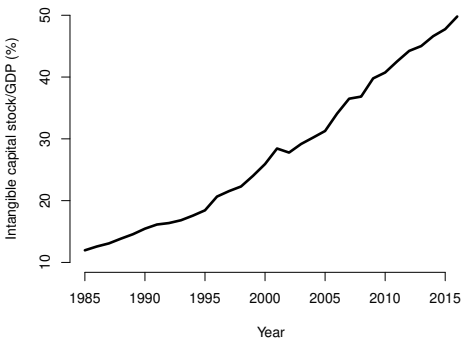
Figure A.1: Time series of aggregate variables.



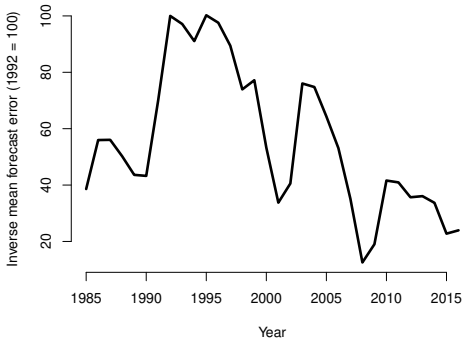
(a) Number of listed firms



(b) Listed firms as share of all firms



(c) Intangible/GDP



(d) Inverse forecast error

Notes: This figure plots the time series of the number (panel A.1a) and share (panel A.1b) of listed firms, intangible capital (panel A.1c), and the inverse forecast error (panel A.1d) in the U.S. The data is from Compustat, I/B/E/S, the World Development Indicators, and the Census Bureau's Business Dynamics Statistics (the all-firm denominator in panel A.1b).

In this section, we explain how we measure the intangible capital stock of public firms. We use firm level data on public U.S. firms from Compustat covering the period from 1985 to 2016 to measure firm-level intangible capital stock. Our baseline measure of *internally* generated intangible capital is the sum of two components: (i) estimated knowledge capital, calculated using research and development expenditure

(XRD); and (ii) estimated organizational capital, calculated using selling, general, and administrative expenses (XSGA). The measure is constructed using the perpetual inventory method, which aggregates net investment flows over the life of the firm:

$$\begin{aligned} \text{[Knowledge capital]}: \quad & k^G_{i,t} = (1-\delta_G)\, k^G_{i,t-1} + R\&D_{it}, \\ \text{[Organizational capital]}: \quad & k^O_{i,t} = (1-\delta_O)\, k^O_{i,t-1} + \gamma_O SG\&A_{it}, \end{aligned}$$

where $R\&D$ is research and development expenditure; $SG\&A$ is selling, general, and administrative expenses. All the intangible flow variables are deflated by the price of intellectual property products from National Income and Product Accounts data (NIPA Table 1.1.9, line 12). δ_G and δ_O are the depreciation rates.[1] γ_O is the fraction of selling,general, and administrative ($SG\&A$) expenditure that adds to the intangible capital stock. We assume $\gamma_O = 0.20$ following Falato et al. (2022). All the empirical results are robust over other reasonable choices of this parameter level.

Then, we calculate the net change in the *acquired* amount of intangibles from changes in the book values of intangibles after the amortization, using Compustat variables INTAN and AM. We obtain the acquired intangible stock $k^B_{i,t}$, applying the perpetual inventory method to the deflated net change in the intangibles.

Our final measure of firm-level intangible capital stock $k^I_{i,t}$ is obtained by combining the internally generated intangible stocks and the acquired intangibles stocks:

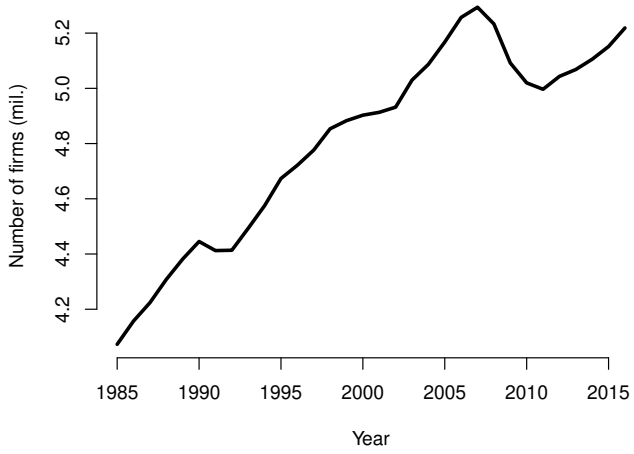
$$k^I_{i,t} = k^G_{i,t} + k^O_{i,t} + k^B_{i,t}$$

[1] We use $\delta_G = \delta_O = 0.15$, which is around the levels used in the literature (Eisfeldt and Papanikolaou, 2013; Falato et al., 2022).

B Industry-level analysis and additional figures

Figure B.1 plots the time series of the total number of firms in the United States. In contrast to the declining trend of the number of listed firms, the total number of firms has been steadily rising.

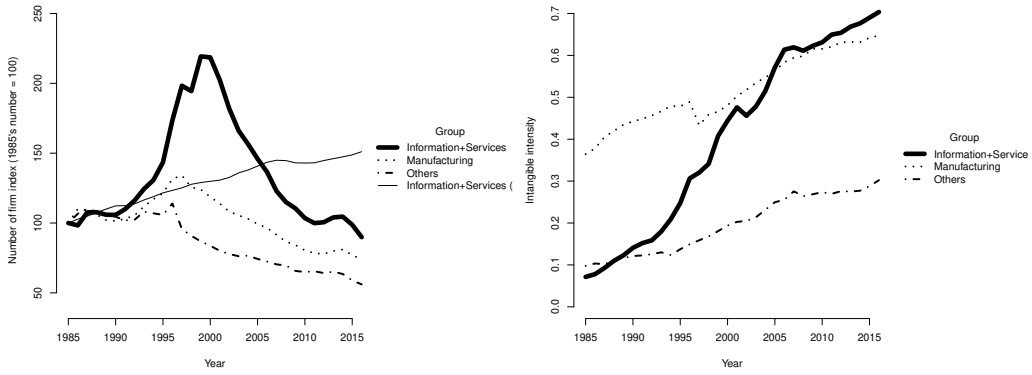
Figure B.1: Number of all firms



Notes: This figure plots the time series of the number of all firms using the Business Dynamics Statistics (BDS) data from the U.S. Census Bureau.

Panel B.2a shows the trend in the number of listed firms for the Information and Services sector, excluding trade and transportation, Manufacturing, and other sectors (Trade and transportation, Agriculture and Mining, Construction). All sectors show an initial increase and then a decline after the mid-nineties, and the decline is much more pronounced in the information and service sector. We also plot the normalized number of all non-listed firms in the information and service sector: as can be seen, only listed firms are affected by the large decline during the entire period of 2000 and 2010 (well after the dot-com bubble), while the overall number is only slightly affected by the 2001 and 2008 recessions. Panel B.2b shows the *intangible intensity*, defined as the ratio of intangible asset to total intangible and tangible asset values,

Figure B.2: Number of listed firms and intangible intensity by industry.



(a) Number of listed firms (b) Intangibles intensity

Notes: This figure shows the trend in the number of listed firms and intangible capital intensity in the U.S. Intangible intensity is defined as the ratio of intangible asset to total intangible and tangible asset values. The groups are defined as Information and Services, excluding trade and transportation, Manufacturing,and other sectors (Trade and transportation, Agriculture and Mining, Construction). Data comes from Compustat.

for the same industries. Manufacturing had historically a higher intangible intensity, which has been taken over in the early 2000s by the service sector.

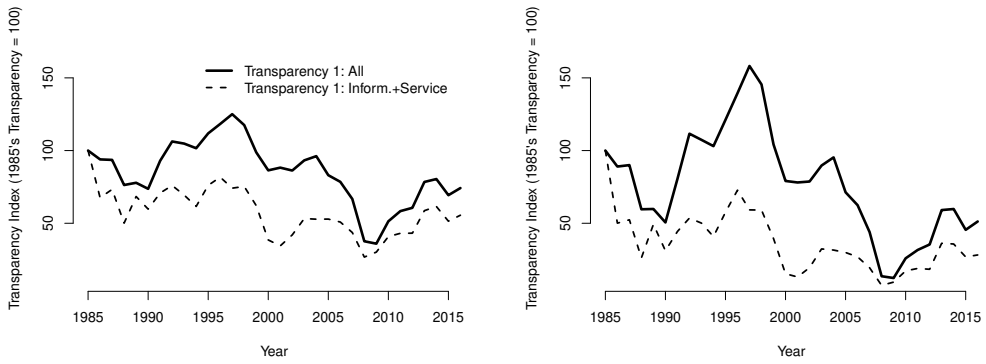
Finally, Figure B.3 shows our transparency measures for the information and service sector, compared to all industries. The information and service sector has a lower transparency over the entire period, and both time series of transparency for all sectors have also declined over time.

Two main take-aways can be taken by the analysis of industry trends: the information and services sector has seen the largest increase in its intangible intensity, and at the same time the largest decline in the number of listed firms and in transparency.

We show the trends for more disaggregated industries in figures B.4 and B.5. Specifically, we report the trends for (a) natural resources and mining, (b) construction, (c) manufacturing, trade, (d) transportation, and utilities, (e) information, (f) professional and business services, (g) education and health services, and (h) leisure and hospitality industries.

Finally, we also report the trends in intangible capital using internally generated

Figure B.3: Time series of transparency for information and service industries.



(a) Transparency 1 (b) Transparency 2

Notes: This figure shows the trend in transparency for information and service industries compared to all industries. Information and Services excludes trade and transportation. Data comes from Compustat and I/B/E/S. See Appendix A for details on measurement.

R&D only (Figure B.6), so that the numbers on intangible intensity can be compared to the ones available in the Bureau of Economic Analysis (BEA), and the time series of the transparency measures for all firms and for only survivors (Figure B.7). Here, the survivor is ex-post conditioned by the firms of which observations are available at the end of the sample period.

Figure B.4: Trends in the number of public firms by industry

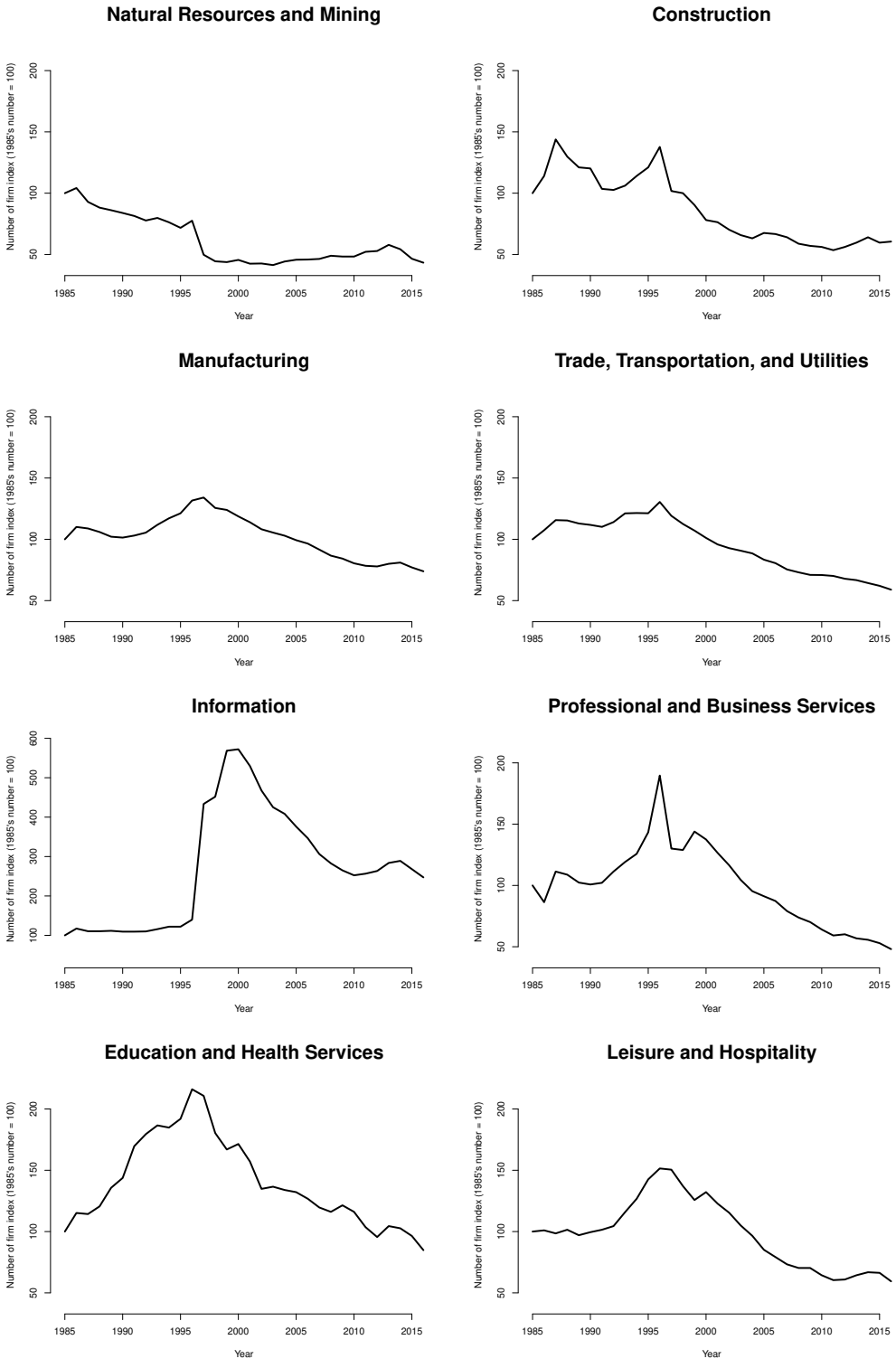


Figure B.5: Trends in intangible intensity by industry

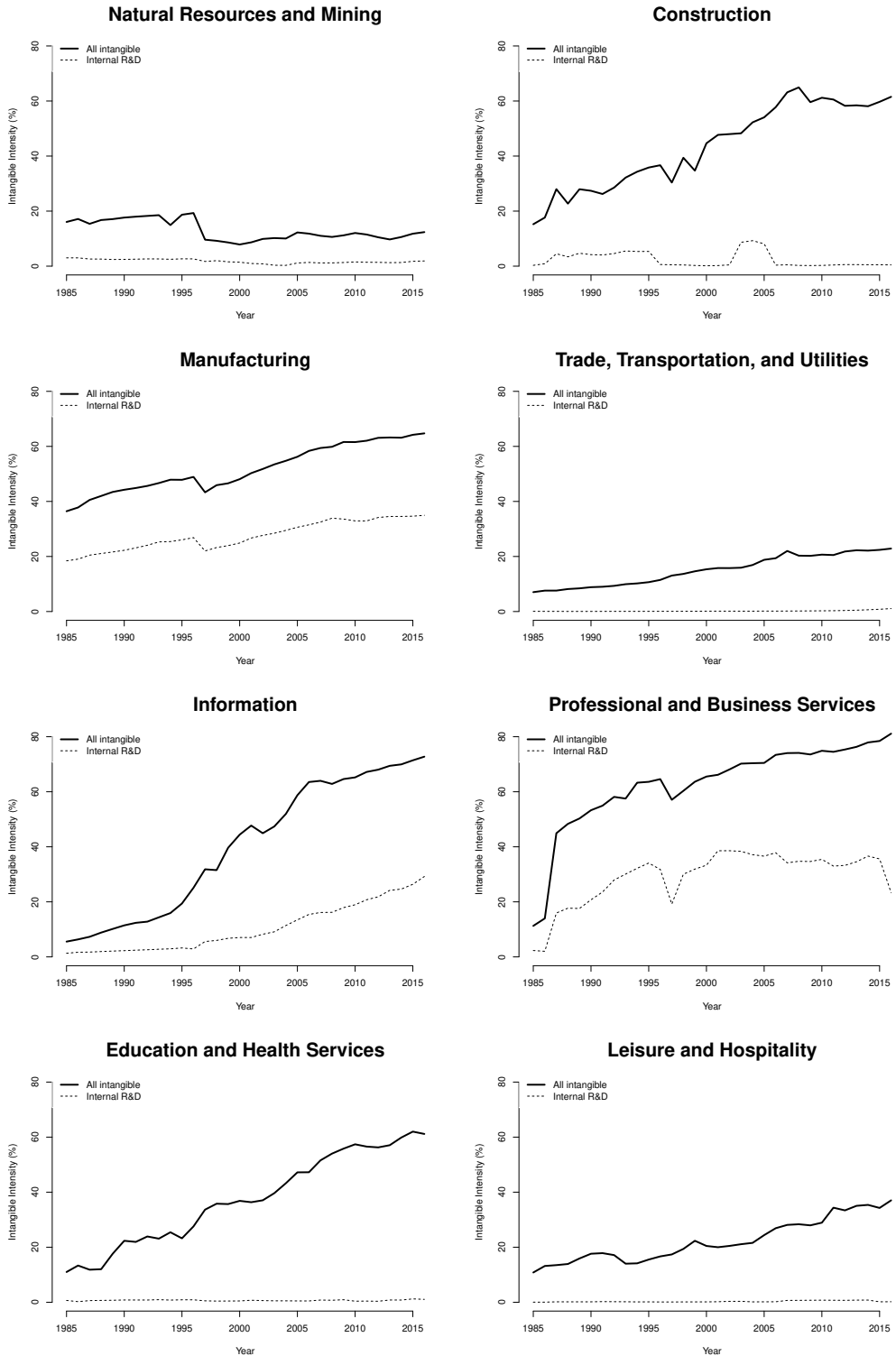
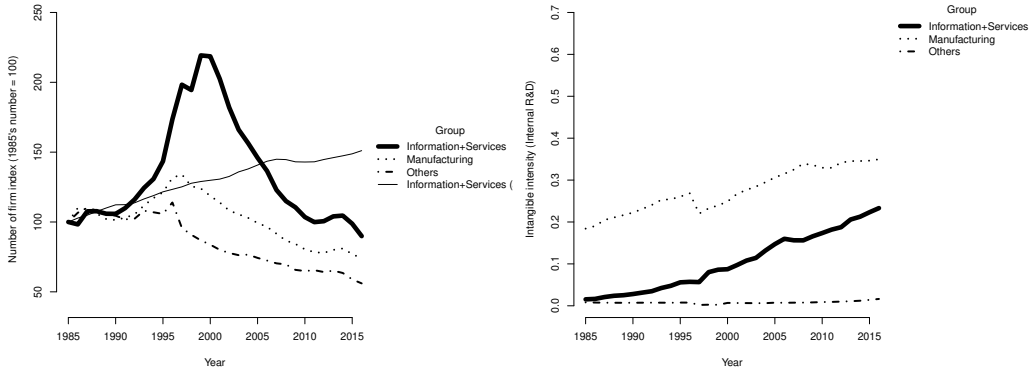


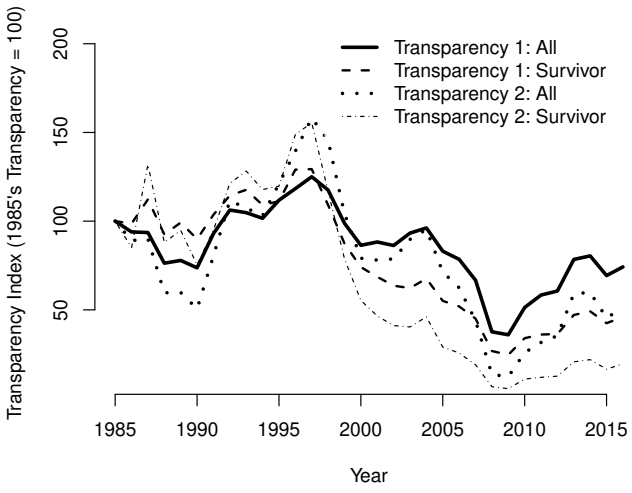
Figure B.6: Number of listed firms and intangible intensity by industry (internal R&D only).



(a) Number of listed firms (b) Intangibles intensity

Notes: This figure shows the trend in the number of listed firms and intangible capital intensity in the U.S. Intangible intensity is defined as the ratio of intangible asset, excluding acquired intangible and organizational capital, to total intangible (again excluding acquired intangible and organizational capital) and tangible asset values.The groups are defined as Information and Services, excluding trade and transportation, Manufacturing,and other sectors (Trade and transportation, Agriculture and Mining, Construction). Data comes from Compustat. See Appendix A for details on measurement.

Figure B.7: Time series of transparency: all firms vs. survivors



Notes: This figure shows the trend in transparency for all firms vs. survivors. Data comes from Compustat and I/B/E/S. See Appendix A for details on measurement.

C Industry-level listing, entry, and exit

The figures in the previous section document qualitatively that the sectors that rely most heavily on intangible capital are also the sectors in which the number of public firms has declined the most. In this section, we complement that evidence with regressions on an industry-year panel, asking whether movements in intangible intensity are associated with movements in the listing rate and in the entry and exit margins of public firms. Recent empirical evidence shows that firms with a higher share of intangible capital have a significantly lower probability of being publicly listed, controlling for other firm characteristics (Celentano and Rempel, 2026). This evidence is especially valuable because it relies on firm-level data that cover private firms as well. The analysis in this section is more descriptive, but its results are consistent with this finding. A second question is whether opacity itself predicts the disappearance of public firms. The model links disclosure to the value of a public listing, so more opaque sectors should, other things equal, lose public firms on net. Accordingly, we also relate the entry, exit, and net exit rates of public firms within an industry and year to the sector's lagged opacity.

We construct a sector-year panel covering 17 two-digit NAICS sectors from 1985 to 2016, excluding the financial sector as in the firm-level analysis. The listing rate is the number of listed firms in Compustat divided by the total number of employer firms in the sector from the Business Dynamics Statistics of the U.S. Census Bureau. The entry (exit) rate is the number of firms whose first (last) Compustat observation falls in year t, divided by the number of listed firms in the sector in year $t-1$; the net exit rate is the difference between the two. Intangible density is aggregate intangible capital over the sum of intangible and physical capital among the listed firms of a sector-year, the same object plotted in Figure B.2; the R&D-only variant uses knowledge capital alone. Sector-level opacity is the median across firms of the firm-year median absolute earnings surprise of Section 4, and it enters the regressions with a one-year lag: opacity measured over the four quarterly announcements of year

$t-1$ predicts entry and exit in year t.[2] All regressions include sector and year fixed effects, and standard errors are clustered by sector.

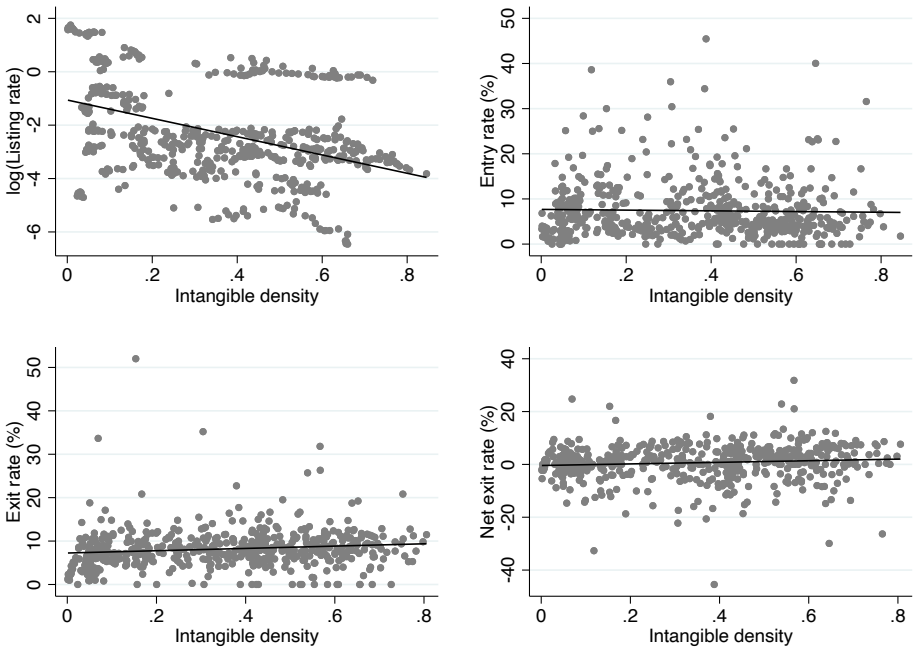
Figures C.1 and C.2 plot the pooled sector-year data. The listing rate declines visibly with intangible density across sectors: this is the between-sector variation summarized by the industry figures above, and it is absorbed by the sector fixed effects in the regressions.

Tables C.1 and C.2 report the regressions for intangible density, weighting each sector-year by the total number of firms in the sector. Within a sector, an increase in intangible density is associated with a lower listing rate, fewer entries, more exits, and a higher net exit rate. A one standard deviation increase in within-sector intangible density (0.12) is associated with an entry rate 0.5 percentage points lower, an exit rate 1.1 percentage points higher, and a net exit rate 1.5 percentage points higher, a large change relative to the sample average net exit rate of 0.6 percent per year. The pattern is similar, and somewhat stronger, when density is measured with R&D-based knowledge capital alone. However, apart from the exit rate, and the net exit rate in the R&D-only specification, these coefficients are imprecisely estimated. Table C.3 reports the corresponding regressions for lagged opacity. More opaque sector-years are followed by a lower listing rate, fewer entries, and more exits; a one standard deviation increase in within-sector lagged opacity is associated with a net exit rate 0.65 percentage points higher, about the size of the average net exit rate; here the net exit coefficient is significant at the 5 percent level.

We interpret these results as descriptive, given the sector-level variation and the small number of clusters, but they line up with the participation margin of the model: sectors that become more intangible-intensive and more opaque lose public firms on net.

[2] One sector-year cell (education in 1989), whose opacity is measured from only two covered firms, is an extreme outlier and is excluded. Results are robust to its inclusion.

Figure C.1: Listing, entry, and exit rates and intangible density.



Notes: Each dot is a sector-year (17 two-digit NAICS sectors, 1985–2016, finance excluded). The listing rate is the number of Compustat listed firms over the number of all employer firms in the sector (Business Dynamics Statistics). Entry, exit, and net exit rates are relative to the number of listed firms in the previous year, in percent; exits are measured through 2015. Intangible density is aggregate intangible capital over intangible plus physical capital among listed firms. Data come from Compustat and the U.S. Census Bureau.

Table C.1: Industry-year regressions: intangible density

	(1) log(Listing rate)	(2) Entry rate	(3) Exit rate	(4) Net exit rate
Intangible density	-0.171	-4.055	8.997**	12.836
	(0.704)	(9.815)	(3.403)	(9.053)
Sector FE	✓	✓	✓	✓
Year FE	✓	✓	✓	✓
Adj. R^2	0.947	0.535	0.168	0.459
Observations	544	544	527	527

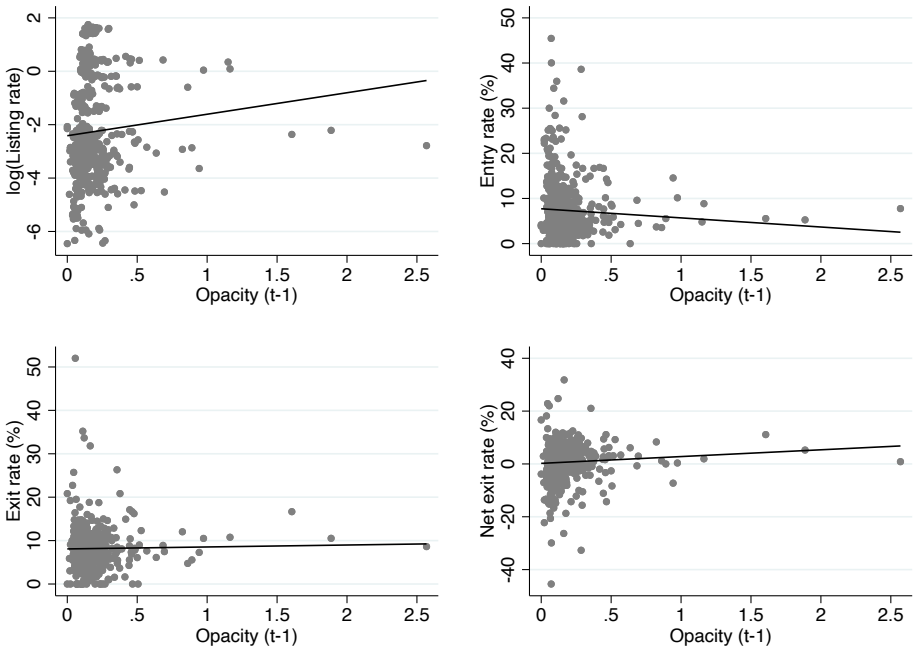
Notes: This table reports regressions of the listing, entry, exit, and net exit rates on intangible density in the sector-year panel (17 two-digit NAICS sectors, 1985–2016). All regressions include sector and calendar year fixed effects and weight each sector-year by the total number of employer firms in the sector from the Business Dynamics Statistics. Exits are measured through 2015. Standard errors (in parentheses) are clustered by sector; *, **, and *** denote significance at the 10, 5, and 1 percent levels.

Table C.2: Industry-year regressions: intangible density, R&D only

	(1) log(Listing rate)	(2) Entry rate	(3) Exit rate	(4) Net exit rate
Intangible density (R&D only)	-1.368	-3.681	10.714*	14.967*
	(1.167)	(7.550)	(5.581)	(7.637)
Sector FE	✓	✓	✓	✓
Year FE	✓	✓	✓	✓
Adj. R^2	0.949	0.534	0.164	0.456
Observations	544	544	527	527

Notes: This table repeats the specification of Table C.1 with intangible density measured from R&D-based knowledge capital only, excluding organizational capital and acquired intangibles.

Figure C.2: Listing, entry, and exit rates and lagged opacity.



Notes: Each dot is a sector-year (17 two-digit NAICS sectors, 1985–2016, finance excluded). Opacity is the sector median across firms of the firm-year median absolute earnings surprise (×100), lagged one year. Outcome definitions as in Figure C.1. Data come from Compustat, I/B/E/S, and the U.S. Census Bureau.

Table C.3: Industry-year regressions: lagged opacity

	(1) log(Listing rate)	(2) Entry rate	(3) Exit rate	(4) Net exit rate
Opacity (t-1)	-0.335	-1.604	1.804	3.538**
	(0.216)	(1.908)	(1.040)	(1.556)
Sector FE	✓	✓	✓	✓
Year FE	✓	✓	✓	✓
Adj. R^2	0.924	0.447	0.187	0.325
Observations	542	542	525	525

Notes: This table reports regressions of the listing, entry, exit, and net exit rates on lagged sector opacity in the sector-year panel (17 two-digit NAICS sectors, 1985–2016). Opacity is the sector median across firms of the firm-year median absolute earnings surprise (×100) in year $t-1$. All regressions include sector and calendar year fixed effects; exits are measured through 2015. Standard errors (in parentheses) are clustered by sector; *, **, and *** denote significance at the 10, 5, and 1 percent levels.

D Summary statistics and full tables

Table D.1: Descriptive Statistics

	Mean	S.D.	P25	P75	N
log(1/med|ES|)	1.908	1.679	0.923	3.019	107,623
log(1/var(ES))	5.217	3.031	3.466	7.178	57,983
Intangible/assets	0.437	0.477	0.128	0.619	107,623
R&D knowledge/assets	0.140	0.370	0.000	0.155	107,623
Organizational/assets	0.175	0.195	0.044	0.242	107,623
Acquired intangibles/assets	0.123	0.176	0.000	0.181	107,623
log(sales)	7.262	1.823	6.117	8.510	107,623
log(employment)	1.565	1.803	0.404	2.791	107,623
Liquid capital/assets	0.432	0.232	0.233	0.610	107,623
Liabilities/assets	0.527	0.211	0.382	0.669	107,623
Book-to-market	0.598	0.518	0.292	0.774	107,623
Age	25.006	16.374	11.000	37.000	107,623
Sales-growth volatility (5y)	0.191	0.245	0.066	0.229	107,623
Number of analysts	3.082	3.356	1.000	4.000	107,623

Notes: This table reports descriptive statistics for the firm-quarter regression sample of Section 4 (1985–2016; one observation per firm-quarter; surprises winsorized at the 1st and 99th percentiles within calendar year before proxy construction).

Sample construction and unit of observation

Four implementation details of the regression sample warrant documentation. First, the unit of observation is the firm-quarter: analyst-level forecast records are collapsed into one observation per firm and fiscal quarter before estimation, so that firms with greater analyst coverage are not implicitly overweighted. Second, price-normalized surprises are winsorized within calendar year at the 1st and 99th percentiles before the proxies are constructed. Third, the balance-sheet control is the liabilities-to-assets ratio: the total-debt item in our constructed panel is populated only from 1997 onward, and conditioning the sample on its availability would restrict estimation to the post-1997 period. Using the fully populated liabilities ratio preserves the entire 1985–2016 window. Fourth, financial firms (two-digit NAICS 52) are excluded from the regression sample and from the aggregate forecastability series; they account

for about 2 percent of the otherwise-screened firm-quarter observations, and their exclusion leaves all estimates essentially unchanged. A merge-attrition audit that tracks observations through each construction step is available from the corresponding author on request.

Table D.2: Intangible components entered one at a time (common sample)

	R&D alone		Organizational alone		Acquired alone	
	1/var	1/med	1/var	1/med	1/var	1/med
Component/assets	-1.047***	-0.292***	-0.848***	-0.625***	1.501***	0.583***
	(0.148)	(0.049)	(0.247)	(0.090)	(0.236)	(0.109)
Observations	57,983	107,623	57,983	107,623	57,983	107,623

Notes: This table reports regressions in which each pair of columns enters a single intangible component, scaled by assets, with the full control set of the main-text regression table (Section 4), estimated on the same firm-quarter samples as the joint-component specification. The dependent variables are the log inverse variance of earnings surprises (1/var) and the log inverse median absolute surprise (1/med). All regressions include industry and calendar year-quarter fixed effects, and standard errors are two-way clustered by firm and year-quarter. The pairwise correlations among the three component ratios are below 0.16, and the one-at-a-time estimates are close to the joint estimates.

Table D.3: Fixed-effect and aggregation robustness

	Industry×year FE		Firm FE		Firm-year collapse	
	1/var	1/med	1/var	1/med	1/var	1/med
Intangible/assets	-0.324***	-0.199***	-1.082***	-0.256***	-0.574***	-0.224***
	(0.115)	(0.035)	(0.134)	(0.043)	(0.098)	(0.028)
Observations	57,983	107,623	57,983	107,623	28,196	47,365

Notes: This table reports the baseline coefficient on the combined intangible-to-assets ratio under alternative specifications: industry × fiscal-year fixed effects (with calendar year-quarter effects), firm fixed effects (with calendar year-quarter effects), and outcomes collapsed to the firm-year level (industry and fiscal-year fixed effects, clustered by firm and fiscal year). The controls are those of the main-text regression table (Section 4).

Table D.4: Subsample results: cohort, long-lived firms, and asset weighting

	Cohort 92–96 & 12–16		Long-lived (≥15y)		Asset-weighted	
	1/var	1/med	1/var	1/med	1/var	1/med
Intangible/assets	0.216	0.017	0.037	-0.103	0.171	0.100
	(0.357)	(0.127)	(0.280)	(0.091)	(0.318)	(0.160)
Observations	8,652	14,460	29,086	60,033	57,983	107,623

Notes: This table reports the baseline specification of the main-text regression table (Section 4) estimated on (i) the cohort of firms observed in both the 1992–1996 and 2012–2016 windows, using observations from those windows only; (ii) the subsample of firms observed for at least fifteen sample years, a long-lived-firm subsample rather than a balanced panel, since conditioning on survival can remove firms near the listing and exit margins; and (iii) the full sample weighted by total assets. The cross-sectional association is statistically indistinguishable from zero in all three cases, indicating that it is concentrated among smaller and shorter-lived firms.

E Standard & Poor's transparency and disclosure

The S&P Transparency and Disclosure index is constructed from a checklist of 98 individual items identified as relevant for assessing firms' transparency, using information drawn from annual reports, Form 10-K filings, and proxy statements. The items are grouped into three broad categories: (i) *ownership structure and investor rights*, (ii) *financial transparency and information disclosure*, and (iii) *board and management structure and process*. Each item is coded on a largely binary basis (1 if the information is disclosed, 0 otherwise). The overall score reflects the ratio of disclosed attributes to the total of 98, and separate scores are calculated for each of the three subcategories. A full description of the methodology, along with the list of the 98 disclosure items, is provided in Appendix 2 and 3 in Patel and Dallas (2002).

For our analysis, the *financial transparency and information disclosure* category is the most relevant. Examples of items in this category include: "Is there a discussion of corporate strategy?", "[does the company] Report details of the products or services produced/provided?", "Does the company disclose its plans for investment in the coming years?", and so on. In contrast, the ownership structure and investor rights and the board and management structure scores are mostly related to governance transparency (examples: "Does the company disclose the voting rights for each class of shares?", "Are specifics of directors' salaries disclosed (numbers)?" and so on).

S&P calculated individual scores for S&P 500 constituents in both June and September 2002, excluding firms for which the S&P had incomplete information (a total of 460 companies). These scores are reported in Appendix 4 of Patel and Dallas (2002). We scrape the table and merge the scores with our dataset using company names, manually verifying all matches to avoid erroneous merges. After the merge, we obtain 328 firms that are matched and have nonmissing information on intangible capital and all covariates.

We then estimate a regression similar to the one in Section 4:

$$\mathrm{y}_i = \alpha + FEs + \beta \times \text{Intangible over total assets}_i + \gamma \times X_i + \varepsilon_i \qquad \text{(E.1)}$$

where y_i is either the overall transparency score or each of the three subcategories score. FEs include industry fixed effects. X_i contains standard firm controls, including book-to-market ratio, liquid capital (cash, inventory, and receivables), leverage (total debt over total assets), log employment, sales (in logs in column (1) and scaled by total assets in columns (2)–(4)), the standard deviation of sales growth over the past three years, and age since IPO. Intangible and liquid capital are scaled by total assets. The control set differs from Section 4 in the leverage measure and the volatility window, and it does not include analyst coverage.

Table E.1 reports the coefficient estimates. We can see there is a negative correlation between intangible capital and the S&P transparency measure, controlling for firm characteristics. The negative association is driven by the financial transparency and information disclosure measure only, consistent with our hypothesis.

Table E.1: Regression of S&P Transparency and Disclosure index on intangibles

	(1) Composite ranking Final	(2) Financial and Information	(3) Ownership	(4) Management and Processes
Intangible	-0.164	-0.517***	0.166	0.135
	(0.107)	(0.125)	(0.209)	(0.114)
Current Assets	-0.439***	-0.778***	-0.505	-0.029
	(0.161)	(0.250)	(0.343)	(0.197)
Leverage	-0.597	0.666	-1.067	-0.956**
	(0.369)	(0.541)	(0.725)	(0.392)
B/M	0.064	-0.040	0.085	0.069
	(0.072)	(0.093)	(0.148)	(0.079)
log(Sales)	-0.002			
	(0.042)			
log(Employment)	0.002	0.012	-0.027	0.008
	(0.038)	(0.033)	(0.052)	(0.026)
Age	0.005***	0.007***	-0.001	0.006***
	(0.002)	(0.003)	(0.003)	(0.002)
sd Past Sales Growth	-0.091	-0.091	0.326	-0.097
	(0.192)	(0.231)	(0.319)	(0.191)
Sales		-0.032	0.042	0.057
		(0.056)	(0.093)	(0.051)
Observations	328	328	328	328

Standard errors in parentheses

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Notes: This table reports the estimates of the coefficients from the following regression

$$\mathrm{y}_i = \alpha + FEs + \beta \times \text{Intangible over total assets}_i + \gamma \times X_i + \varepsilon_i$$

where y_i is either the overall S&P transparency score or each of the three subcategories score. FEs include industry fixed effects. X_i represents firm controls. Standard errors are heteroskedasticity-robust.

F Welfare measure derivation

Here, we derive the money-metric financier-welfare measure in the main text. For each active public cell, let $CE(q)$ denote the certainty equivalent of a local specialized financier, net of policy-invariant outside wealth. Let CE^N denote its private-market counterpart. Each listed submarket is served by a unit mass of financiers, while a mass $1/\nu_N$ of financiers serves the private market; equal welfare weights on financiers therefore imply

$$\begin{aligned} W^F(\overline{q}) &:= \int_0^{1-\overline{q}} CE(q)dq + \frac{1}{\nu_N} CE^N \\ &= \int_0^{1-\overline{q}} \left[x(q)\frac{\pi(q)}{P(q)} - \frac{\Lambda}{2}x(q)^2 \frac{1}{\xi + \psi(\overline{q}+q)} \right] dq + \frac{1}{\nu_N}\left[x^N \frac{\pi^N}{P^N} - \frac{\Lambda}{2}(x^N)^2 \frac{1}{\xi} \right]. \end{aligned}$$

Each financier's first-order condition gives

$$\Lambda x(q) H(q;\overline{q}) = \frac{\pi(q)}{P(q)}, \qquad\qquad \Lambda x^N H(0;0) = \frac{\pi^N}{P^N}.$$

Consequently, the local uncertainty cost equals one half of expected local excess payoff. Substituting these conditions and then using $x(q)/P(q) = \mathcal{M}(q)$ and $x^N/P^N = \nu_N M_N$ yields

$$\begin{aligned} W^F(\overline{q}) &= \frac{1}{2}\int_0^{1-\overline{q}} x(q)\frac{\pi(q)}{P(q)}dq + \frac{1}{2\nu_N} x^N \frac{\pi^N}{P^N} \\ &= \frac{1}{2}\int_0^{1-\overline{q}} \mathcal{M}(q)\pi(q)dq + \frac{1}{2} M_N \pi^N. \end{aligned} \tag{F.1}$$

This is an explicit sum of local money-metric certainty equivalents. It is not a von Neumann–Morgenstern utility for a pooled representative investor. Independent local payoff shocks have zero aggregate risk under the exact law of large numbers, but their certainty-equivalent costs remain because financiers cannot trade across cells or insure one another. The formula is also resource-consistent: aggregate expected financier income, $\int \mathcal{M}\pi\, dq + M_N\pi^N$, equals the operating profits generated by listed and private

firms, and welfare is one half of that income by the envelope property of the certainty equivalent.

G Proofs

G.1 Proof for Proposition 1.

Proposition 1 (Funding supply)**.**

A specialized financier's optimal funding for listed firms with transparency q, $x^*(q)$, and the corresponding private-market funding, x^{N*}, are

$$x^*(q) = \frac{\overline{r}(q)}{\Lambda H(q;\overline{q})} = \frac{\pi(q)/P(q)}{\Lambda/(\xi + \psi(\overline{q}+q))}, \quad x^{N*} = \frac{\pi^N/P^N}{\Lambda/\xi}. \tag{G.1}$$

Proof.

Fix an active public submarket q and one of its specialized local financiers j. The financier cannot trade risky claims outside this cell or insure its local payoff innovation. With a zero-excess-return outside asset, terminal wealth is

$$C_j(q) = a_j(q) + x_j(q)\widetilde{r}_j(q), \qquad \widetilde{r}_j(q) \sim N(\overline{r}(q), H(q;\overline{q}))\,. \tag{G.2}$$

The upper bound $x_j(q) \leq a_j(q)$ is nonbinding by assumption. The normal moment-generating function gives

$$\mathbb{E}\left[-e^{-\Lambda C_j(q)}\right] = -\exp\left\{-\Lambda\left[a_j(q) + x_j(q)\overline{r}(q) - \frac{\Lambda}{2}x_j(q)^2 H(q;\overline{q})\right]\right\}. \tag{G.3}$$

Because the outer transformation is strictly increasing in the bracketed term, the financier equivalently maximizes the local money-metric certainty equivalent

$$CE_j(q) = a_j(q) + x_j(q)\overline{r}(q) - \frac{\Lambda}{2}x_j(q)^2 H(q;\overline{q}). \tag{G.4}$$

For the positive expected returns in equilibrium, the unique interior first-order con-

dition is

$$\overline{r}(q) - \Lambda x_j^*(q) H(q; \overline{q}) = 0. \tag{G.5}$$

Symmetry across the unit mass of financiers in the cell and the definitions of $\overline{r}$ and H therefore imply

$$x^*(q) = \frac{\overline{r}(q)}{\Lambda H(q; \overline{q})} = \frac{\pi(q)/P(q)}{\Lambda/(\xi + \psi(\overline{q} + q))}. \tag{G.6}$$

The same local argument for a private-market financier, using $\overline{r}^N = \pi^N/P^N$ and $H(0;0) = 1/\xi$, gives

$$x^{N*} = \frac{\overline{r}^N}{\Lambda H(0;0)} = \frac{\pi^N/P^N}{\Lambda/\xi}. \tag{G.7}$$

No variance of an aggregate portfolio enters either first-order condition. Independent local innovations wash out under the exact law of large numbers, while segmentation makes each financier bear its own innovation. ■

G.2 Proof for Proposition 2.

Proposition 2. *(Intangibles and the transparency)*
The input demand of intangible capital k_I is as follows:

$$k_I(q, \mathcal{M}; \overline{q}) = \left(\left(\frac{\alpha(\Phi^{ex})^\gamma}{r} \right)^{\frac{1}{1-\alpha-\theta}} \left(\frac{r\theta}{p\alpha} \right)^{\frac{1-\alpha}{1-\alpha-\theta}} \right) (1 - \overline{q} - q)^{\frac{\theta}{1-\alpha-\theta}}. \tag{G.8}$$

Proof.
From the first-order optimality condition,

$$[k_T]: \quad \alpha k_T^{\alpha-1} (k_I(1 - \overline{q} - q))^\theta (\Phi^{ex})^\gamma = r \tag{G.9}$$

$$[k_I]: \quad \theta k_T^\alpha (k_I(1 - \overline{q} - q))^{\theta-1} (\Phi^{ex})^\gamma (1 - \overline{q} - q) = p \tag{G.10}$$

From the first-order conditions with respect to k_T and k_I, we obtain

$$\frac{r}{p} = \left(\frac{\alpha}{\theta}\right) \frac{k_I}{k_T}. \tag{G.11}$$

Substituting this relation into the first-order condition with respect to k_T, we get

$$r = \alpha \left(\frac{\alpha p}{\theta r}\right)^{\alpha-1} (k_I)^{\alpha+\theta-1} (1 - \overline{q} - q)^{\theta} (\Phi^{ex})^{\gamma}. \tag{G.12}$$

Thus,

$$k_I = \left(\left(\frac{\alpha(\Phi^{ex})^{\gamma}}{r}\right)^{\frac{1}{1-\alpha-\theta}} \left(\frac{r\theta}{p\alpha}\right)^{\frac{1-\alpha}{1-\alpha-\theta}} \right) (1 - \overline{q} - q)^{\frac{\theta}{1-\alpha-\theta}} = A(1 - \overline{q} - q)^{\frac{\theta}{1-\alpha-\theta}}, \tag{G.13}$$

where $A := \left(\left(\frac{\alpha(\Phi^{ex})^{\gamma}}{r}\right)^{\frac{1}{1-\alpha-\theta}} \left(\frac{r\theta}{p\alpha}\right)^{\frac{1-\alpha}{1-\alpha-\theta}} \right)$. The proposition is immediate from the last equation. ■

G.3 Proof for Proposition 3 (via the ODE representation).

Proposition 3. *(Transparency distribution)*
The unnormalized probability density function $\mathcal{M}$ of transparency q has the following analytic form:

$$\mathcal{M}(q) = (\xi + \psi(\overline{q} + q)) (1 - \overline{q} - q)^{\frac{\theta}{1-\alpha-\theta}} \frac{1}{\phi^N} \tag{G.14}$$

where $\phi^N = \frac{\xi}{\nu_N M_N}$.

Proof.

We derive the following equations using the first-order condition with respect to q:[3]

$$\frac{\phi^{L'}(q)}{\phi^{L}(q)} = \frac{\theta k_T^{\alpha}(k_I(1-\overline{q}-q))^{\theta-1}(\Phi^{ex})^{\gamma}k_I}{k_T^{\alpha}(k_I(1-\overline{q}-q))^{\theta}(\Phi^{ex})^{\gamma} - rk_T - pk_I} \tag{G.15}$$

$$= \frac{\theta k_T^{\alpha}(k_I(1-\overline{q}-q))^{\theta-1}(\Phi^{ex})^{\gamma}k_I}{(1-\alpha-\theta)k_T^{\alpha}(k_I(1-\overline{q}-q))^{\theta}(\Phi^{ex})^{\gamma}} \tag{G.16}$$

$$= \frac{\theta}{1-\alpha-\theta}\left(\frac{1}{1-\overline{q}-q}\right). \tag{G.17}$$

From $\frac{\partial}{\partial q}log(\phi^{L}(q)) = \frac{\phi^{L'}(q)}{\phi^{L}(q)}$, we guess and verify that the solution of the first-order differential equation takes the following form:

$$\phi^{L}(q) = (1-\overline{q}-q)^{n}\,\widetilde{C}, \tag{G.18}$$

for some $n \in \mathbb{R}$ and some $\widetilde{C} \in \mathbb{R}$. From the indifference condition in the equilibrium, $\pi(q)\phi^{L}(q)$ does not depend on q.

$$\pi(q)\phi^{L}(q) = \left((1-\alpha-\theta)\left(\frac{\alpha p}{\theta r}\right)^{\alpha} A^{\alpha+\theta}(1-\overline{q}-q)^{\frac{\theta}{1-\alpha-\theta}}\Phi^{\gamma}\right)(1-\overline{q}-q)^{n}\,\widetilde{C}. \tag{G.19}$$

where $A := \left(\left(\frac{\alpha(\Phi^{ex})^{\gamma}}{r}\right)^{\frac{1}{1-\alpha-\theta}}\left(\frac{r\theta}{p\alpha}\right)^{\frac{1-\alpha}{1-\alpha-\theta}}\right)$.

Therefore,

$$n = -\frac{\theta}{1-\alpha-\theta}. \tag{G.20}$$

This leads to $\phi^{L}(q) = (1-\overline{q}-q)^{-\frac{\theta}{1-\alpha-\theta}}\,\widetilde{C}$.

[3]Here the proof is based on the first-order conditions that are simultaneously obtained for k_T, k_I, and q for the brevity of notations. The optimal allocations stay unaffected in this problem even if the solution is solved sequentially (interim problem first (k_T and k_I), and then q).

Then, the distribution of listed firms is as follows:

$$\mathcal{M}(q) = (\xi + \psi(\overline{q} + q))/\phi^L(q) \tag{G.21}$$

$$= (\xi + \psi(\overline{q} + q))\,(1 - \overline{q} - q)^{\frac{\theta}{1-\alpha-\theta}}\,\frac{1}{\widetilde{C}}. \tag{G.22}$$

From the indifference condition between listed and non-listed,

$$\phi^N = \frac{\pi(q)\phi^L(q)}{\pi^N} \tag{G.23}$$

$$= \frac{\left((1-\alpha-\theta)\left(\frac{\alpha p}{\theta r}\right)^{\alpha} A^{\alpha+\theta}(1-\overline{q}-q)^{\frac{\theta}{1-\alpha-\theta}}\Phi^{\gamma}\right)(1-\overline{q}-q)^{-\frac{\theta}{1-\alpha-\theta}}\,\widetilde{C}}{\left((1-\alpha-\theta)\left(\frac{\alpha p}{\theta r}\right)^{\alpha} A^{\alpha+\theta}\Phi^{\gamma}\right)} \tag{G.24}$$

$$= \widetilde{C}. \tag{G.25}$$

Therefore, $\mathcal{M}(q) = (\xi + \psi(\overline{q} + q))\,(1 - \overline{q} - q)^{\frac{\theta}{1-\alpha-\theta}}\,\frac{1}{\phi^N}$.

In the equilibrium, $\phi^N (= \widetilde{C})$ is determined at the level where the following equation holds:

$$\int_0^{1-\overline{q}} \mathcal{M}(q)dq = 1 - M_N.$$

■

G.4 Proof for Corollary 3.

Corollary 3. *(Truncated normalized Beta distribution)*
The gross transparency, $y := q+\overline{q}$, is a shifted and scaled transformation of a random variable Y that follows a truncated $Beta(B+1,2)$ distribution:

$$q + \overline{q} = 1 - (1 + \xi/\psi) \times Y \tag{G.26}$$

where Y has the $Beta(B+1,2)$ density truncated to $[0, \frac{1-\overline{q}}{1+\xi/\psi}]$ and renormalized to integrate to one, $B := \theta/(1-\alpha-\theta)$, and the unnormalized firm measure $\mathcal{M}$ equals $(1-M_N)$ times the density of q induced by this distribution through the transformation

$q = 1 - \overline{q} - (1 + \xi/\psi)Y$.

Proof.

We define $M_Y(y)$ as the unnormalized measure over the random variable $Y = \frac{1-q-\overline{q}}{1+\xi/\psi}$ induced by the firm measure $\mathcal{M}$. Note that $q + \overline{q} = 1 - (1 + \xi/\psi)Y$. Then,

$$M_Y(y) \propto (1 - y)y^B \quad \text{for} \quad y \in \left[0, \frac{1 - \overline{q}}{1 + \xi/\psi}\right], \tag{G.27}$$

with total mass $\int_0^{\frac{1-\overline{q}}{1+\xi/\psi}} M_Y(y)dy = 1 - M_N$. The kernel $(1-y)y^B$ is that of a $Beta(B+1, 2)$ density; dividing M_Y by its total mass $1 - M_N$ therefore yields the $Beta(B+1, 2)$ density truncated to $[0, \frac{1-\overline{q}}{1+\xi/\psi}]$ and renormalized to integrate to one, as stated. ■

G.5 Proof for Proposition 4

Proposition 4 (Non-listed firms' measure)**.**

In equilibrium, the measure of non-listed firms M_N is as follows:

$$M_N = \frac{1}{1 + \psi\frac{\nu_N}{\xi}(1 + \frac{\xi}{\psi})^{B+2}\mathcal{B}(B + 1, 2)F\left(\frac{1-\overline{q}}{1+\xi/\psi}; B + 1, 2\right)} \tag{G.28}$$

where $\mathcal{B}$ is the beta function, and F is the cumulative distribution function of beta distribution.[4]

Proof. We have the following closed-form solution for M_N:

$$M_N = \frac{1}{1 + \psi\frac{\nu_N}{\xi}\int_0^{1-\overline{q}}(\frac{\xi}{\psi} + (\overline{q} + q))(1 - \overline{q} - q)^B dq}. \tag{G.30}$$

Using Corollary 3, we can integrate out the $M(q)$ in the right-hand side of the

[4]The beta function is defined as follows:

$$\mathcal{B}(a, b) := \frac{\Gamma(a)\Gamma(b)}{\Gamma(a + b)} = \int_0^1 x^{a-1}(1 - x)^{b-1}dx. \tag{G.29}$$

equation in the following steps, using $y = \frac{1-q-\overline{q}}{1+\xi/\psi} \in [0, \frac{1-\overline{q}}{1+\xi/\psi}]$:

$$M_N = \frac{1}{1 + \psi\frac{\nu_N}{\xi}(1+\frac{\xi}{\psi})^{B+2}\int_0^{\frac{1-\overline{q}}{1+\xi/\psi}}(1-y)(y)^B dy}. \tag{G.31}$$

$$M_N = \frac{1/\mathcal{B}(B+1,2)}{1/\mathcal{B}(B+1,2) + \psi\frac{\nu_N}{\xi}(1+\frac{\xi}{\psi})^{B+2}\frac{1}{\mathcal{B}(B+1,2)}\int_0^{\frac{1-\overline{q}}{1+\xi/\psi}} y^B(1-y)dy}. \tag{G.32}$$

We integrate the denominator using the cumulative distribution function of beta distribution, F:

$$M_N = \frac{1/\mathcal{B}(B+1,2)}{1/\mathcal{B}(B+1,2) + \psi\frac{\nu_N}{\xi}(1+\frac{\xi}{\psi})^{B+2}F\left(\frac{1-\overline{q}}{1+\xi/\psi}; B+1, 2\right)}. \tag{G.33}$$

By multiplying $\mathcal{B}(B+1,2)$ on the numerator and the denominator, we obtain the following analytic form:

$$M_N = \frac{1}{1 + \psi\frac{\nu_N}{\xi}(1+\frac{\xi}{\psi})^{B+2}\mathcal{B}(B+1,2)F\left(\frac{1-\overline{q}}{1+\xi/\psi}; B+1, 2\right)}. \tag{G.34}$$

■

G.6 Proof for Proposition 5.

Proposition 5 (Drivers of listing decisions)**.**
For $\overline{q} \in (0,1)$ and $\theta, \psi, \nu_N, \xi > 0$, the equilibrium measure of non-listed firms M_N strictly

(i) increases in $\overline{q}$ (mandated transparency),

(ii) increases in θ (intangible share),

(iii) decreases in ψ (transparency's information content),

(iv) decreases in ν_N (limited private-market participation),

(v) increases in ξ (baseline information level).

Proof.

i) We have

$$M_N = \frac{1}{1+\psi\frac{\nu_N}{\xi}(1+\frac{\xi}{\psi})^{B+2}\mathcal{B}(B+1,2)F\left(\frac{1-\overline{q}}{1+\xi/\psi};B+1,2\right)}. \tag{G.35}$$

F decreases in $\overline{q}$, and M_N decreases in F. Thus, M_N increases in $\overline{q}$.

For the other results, we write M_N in the following form:

$$M_N = \frac{1}{1+\psi\frac{\nu_N}{\xi}\int_0^{1-\overline{q}}(\frac{\xi}{\psi}+(\overline{q}+q))(1-\overline{q}-q)^B dq.} \tag{G.36}$$

ii) Taking the partial derivative with respect to θ, we get

$$\frac{\partial M_N}{\partial\theta} = \underbrace{-(M_N)^2}_{<0}\left(\underbrace{\psi\frac{\nu_N}{\xi}\int_0^{1-\overline{q}}(\frac{\xi}{\psi}+(\overline{q}+q))(1-\overline{q}-q)^B\overbrace{\frac{\partial B}{\partial\theta}}^{>0}\overbrace{log(1-q-\overline{q})}^{<0}dq}_{<0}\right) \tag{G.37}$$

$$> 0. \tag{G.38}$$

iii) Taking the partial derivative with respect to ψ, we get

$$\frac{\partial M_N}{\partial\psi} = \underbrace{-(M_N)^2}_{<0}\left(\underbrace{\frac{\nu_N}{\xi}\int_0^{1-\overline{q}}(\overline{q}+q)(1-\overline{q}-q)^B dq}_{>0}\right) \tag{G.39}$$

$$< 0. \tag{G.40}$$

iv) Taking the partial derivative with respect to ν_N, we get

$$\frac{\partial M_N}{\partial \nu_N} = \underbrace{-(M_N)^2}_{<0} \left(\underbrace{\frac{\psi}{\xi} \int_0^{1-\overline{q}} (\frac{\xi}{\psi} + (\overline{q}+q))(1-\overline{q}-q)^B dq}_{>0} \right) \tag{G.41}$$

$$< 0. \tag{G.42}$$

v) Taking the partial derivative with respect to ξ, we get

$$\frac{\partial M_N}{\partial \xi} = \underbrace{-(M_N)^2}_{<0} \left(\underbrace{\psi \nu_N \int_0^{1-\overline{q}} (\overline{q}+q) \overbrace{\left(-\frac{1}{\xi^2}\right)}^{<0} (1-\overline{q}-q)^B dq}_{<0} \right) \tag{G.43}$$

$$> 0. \tag{G.44}$$

■

H Measurement augmentation and identification of the information block

The calibration of ξ in Section 5 treats the upper tail of listed-firm forecast dispersion as an out-of-model benchmark rather than a matched moment. This appendix documents why. Within the model, every listed firm's dispersion satisfies $sd(q) = 1/\sqrt{\xi + \psi(\overline{q} + q)} \leq 1/\sqrt{\xi + \psi\overline{q}}$, which equals 12.6 percent at the baseline fit and 28.0 percent at the post-change fit, whereas the observed 99th-percentile dispersion among listed firm-quarters, the calibration benchmark, is far larger. Estimating $(\overline{q}, \psi, \xi)$ jointly against the listed share, the mean dispersion, and a correctly computed listed-tail moment (the 99th percentile of the conditional listed dispersion distribution) is infeasible: matching the listed share forces $\overline{q}$ close to one, the voluntary-disclosure band collapses, and the model-implied tail coincides with the mean.

To ask whether the data can nonetheless separate disclosure-driven and fundamental components, we augment the measurement equation: the observed dispersion variance is $V_i = H(q_i; \overline{q}) + \sigma_i^2$, where σ_i^2 is a firm-level fundamental forecast-difficulty component, lognormally distributed, orthogonal to the disclosure channel, and affecting measurement but not priced in funding. Fitting $(\overline{q}, \psi, \xi)$ and the two σ-distribution parameters to five listed moments (the listed share and the mean, median, 90th percentile, and top-1-percent mean of the firm-quarter distribution of $\sqrt{V_i}$, computed from the revised firm-quarter construction of Section 4, which excludes financial firms and differs in scale from the original proxy) matches the 1992–1996 moments closely and the 2012–2016 moments approximately. The fit, however, attributes only 0.2 percent (1992–1996) and 0.04 percent (2012–2016) of observed dispersion variance to the disclosure component, and the information parameters are not identified: parameter vectors with ψ differing by several orders of magnitude achieve equivalent fit, because once the fundamental component absorbs the dispersion distribution, only the listing

margin restricts $(\overline{q}, \psi, \xi)$.

The cross-sectional dispersion of listed forecast errors therefore cannot identify the disclosure-driven information block: in the augmented model, the information block is not identified. In the restricted baseline model, conditional on the externally calibrated ξ and excluding the added fundamental component, the mean dispersion and the listing share pin down $(\overline{q}, \psi)$. This finding also quantifies, within the structural framework, the limitation noted in Section 4: analyst-based dispersion conflates chosen opacity with fundamental uncertainty, and its cross-sectional variation is dominated by the latter.

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